

DOCUMENT 2:

HARDCORE PROCESSOR V/S SOFTCORE PROCESSOR

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1.Introduction to SoC Architectures

In modern digital design, Field Programmable Gate Arrays (FPGAs) are rarely used as standalone logic devices. Instead, they are integrated with standard microprocessors to create a System-on-Chip (SoC). This allows the sequential, complex decision-making to be handled by a C-programmed processor (Software) and interrupt handling, while heavy, repetitive mathematical tasks (like an FFT) are calculated using physical circuits (Hardware).

In this project, two entirely different processor architectures were evaluated:

- **Hardcore** ARM Cortex-A9
- **Softcore** Xilinx MicroBlaze.

2. The Hardcore Processor (Zynq-7000 SoC)

A "Hardcore" processor is a physical piece of silicon. In the Zynq-7000 architecture, an ARM Cortex-A9 processor is physically manufactured right next to the FPGA fabric on the exact same die.

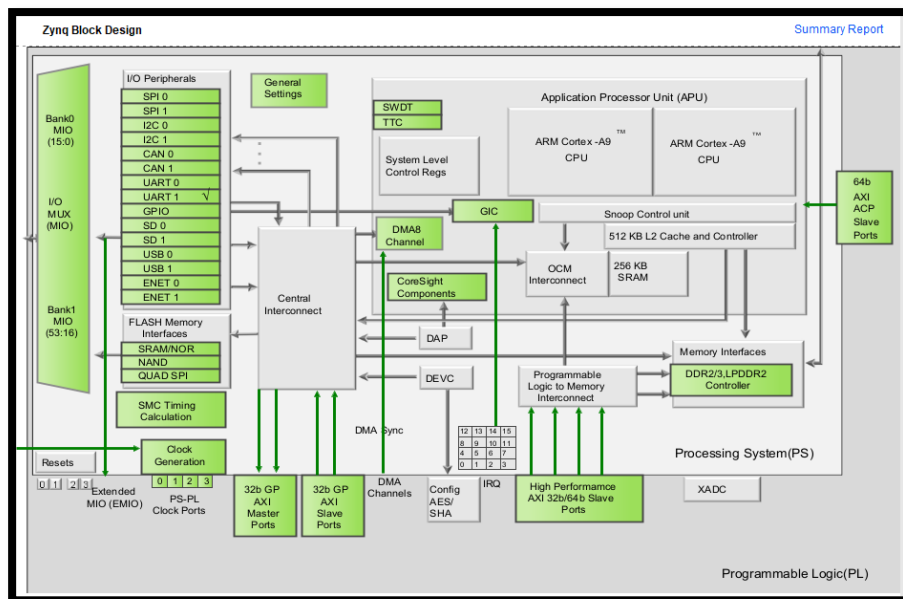
Key Characteristics:

- **Physical Silicon:** Because it is a real physical chip, it cannot be modified, resized, or removed.
- **High Performance:** It runs at significantly higher clock speeds (typically 667 MHz or more) because the silicon paths are permanently optimized.
- **Controllers:** It features a physical, hard-wired memory controller. It does not need the Programmable Logic (PL) fabric to talk to external DDR RAM, saving massive amounts of logic gates.

2.1 The Processing System (PS)

Zynq processor outside the FPGA, on the same chip.

- **The Core:** It features a dual-core ARM Cortex-A9 processor. Because it is physically manufactured into the silicon, it can run at much higher clock speeds (up to 1 GHz) than a soft-core processor.



- **Built-in Peripherals:** Unlike an FPGA where you have to design controllers from scratch, the PS comes with physical hardware controllers already built-in (SPI, I2C, CAN, UART, USB, and Ethernet).
- **Memory Controllers:** It has a dedicated physical DRAM Controller to talk to DDR memory directly, completely bypassing the FPGA logic.

2.2 The Programmable Logic (PL)

The PL is the FPGA side of the chip. The PL is used to build custom hardware.

2.3 The Internal Bridge (PS-PL Interconnect)

The true power of the Zynq lies in how the ARM processor (PS) and the FPGA (PL) talk to each other. They communicate across the silicon boundary using AMBA AXI4 Interconnects.

- **General Purpose (GP) AXI Ports:** Used for basic control. The ARM processor uses these to send low-speed commands to the FPGA (e.g., "turn on that LED" or "start the motor").
- **High Performance (HP) AXI Ports:** These are the massive data highways. If the FPGA captures a 4K video frame, it uses the HP ports (often via a DMA block) to blast that data directly into the ARM processor's DDR memory.
- **EMIO (Extended Multiplexed I/O):** If the ARM processor runs out of physical pins for its peripherals (like needing an extra SPI bus), it can route its internal signals through the EMIO boundary and use the FPGA's pins to reach the outside world.

The Softcore Processor (Xilinx MicroBlaze)

A "Softcore" processor is not a physical chip. The Xilinx Microblaze is a 32-bit RISC processor built entirely out of programmable logic gates (Look-Up Tables, Flip-Flops, and Multiplexers) inside the FPGA fabric.

Key Characteristics:

- **Synthesized RTL:** It is essentially a massive, pre-written hardware design that Vivado synthesizes
- **Highly Customizable:** You can add or remove instructions (like floating-point math units) based on your exact project needs.(for the FFT we requires FPU for math instructions)
- **Lower Clock Speeds:** Because it is routed through programmable fabric rather than permanent silicon pathways, it runs at lower frequencies (typically 100 MHz to 200 MHz).
- **High Resource Cost:** It consumes thousands of logic gates. Furthermore, if it needs to access external DDR RAM, it requires a massive Memory Interface Generator (MIG) IP block to be synthesized alongside it.

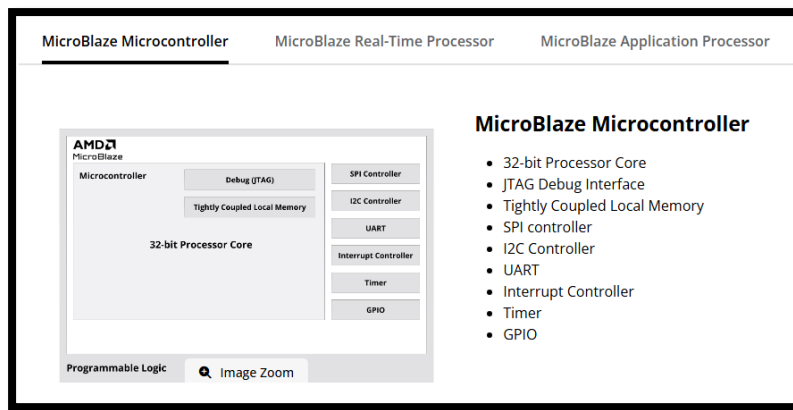
The Microblaze soft-core processor can be scaled from a tiny, logic-efficient controller to a complex, Linux-capable Real time processor. As the system requirements increase, new hardware blocks are added into the FPGA logic (PL).

1. Microblaze Microcontroller

This is the smallest, most efficient footprint. It is designed for simple, deterministic hardware control using "bare-metal" C programming (no operating system).

Included Hardware:

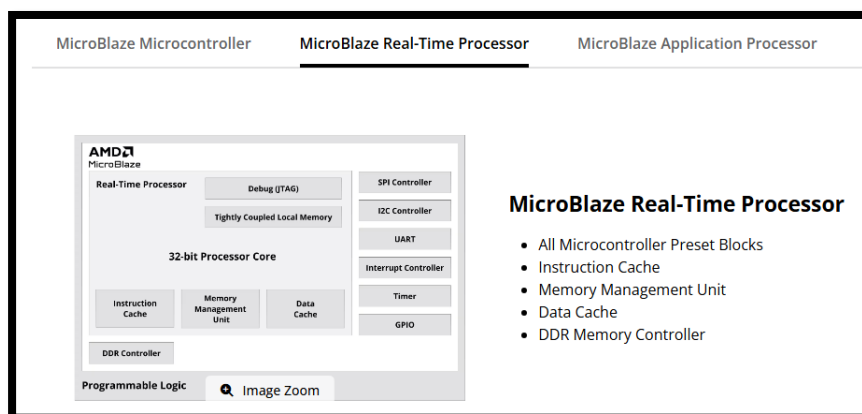
- **32-bit Processor Core:** The central arithmetic and logic brain.
- **Tightly Coupled Local Memory:** This is the ultra-fast BRAM connected via the Local Memory Bus (LMB). It allows the CPU to boot instantly and access variables in a single clock cycle.
- **Standard Peripherals:** SPI, I2C, UART, Interrupt Controller, Timer, and GPIO. These are the basic interfaces needed to talk to sensors, flash memory, terminals, and physical pins.



2. Microblaze Real-Time Processor

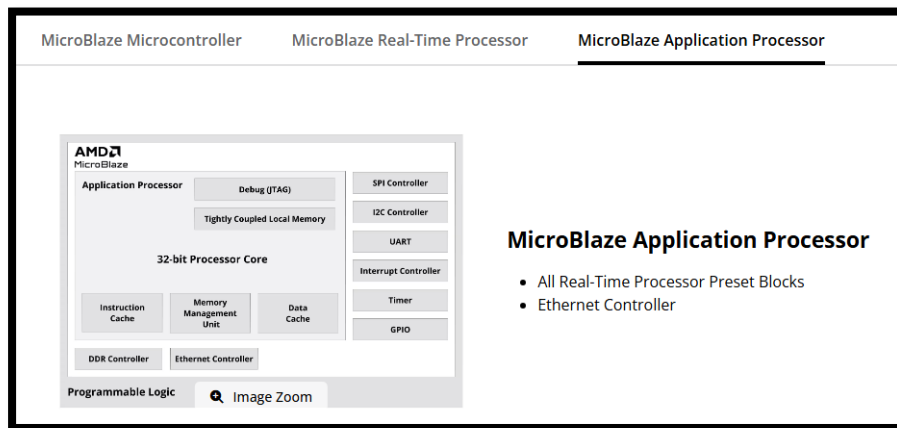
This configuration introduces external memory capabilities and the hardware required to run a Real-Time Operating System (RTOS). It is deterministic.

- **DDR Controller:** In the Microcontroller tier, the CPU only uses local BRAM. This new controller acts as the bridge to massive, off-chip external memory (DDR RAM), allowing the system to store massive data arrays or complex RTOS software.
- **Instruction & Data Caches:** Because external DDR memory is physically far away and has high latency, the CPU cannot read from it directly without stalling. Caches are small pockets of ultra-fast internal memory. The hardware automatically pre-loads chunks of DDR data and upcoming software instructions into these caches so the CPU can execute them at full speed.
- **Memory Management Unit (MMU):** When running an operating system with multiple tasks, tasks can accidentally overwrite each other's memory and crash the system. The MMU provides hardware-level Memory Protection, to stop the overwriting the data to the memory.



3. Microblaze Application Processor

It contains every block from the previous tiers and is specifically designed to boot a full, high-level operating system like Linux.



Architectural Comparison Summary

Hardcore Processor (Zynq ARM)	Softcore Processor (Microblaze)
Dedicated, permanent silicon chip.	Synthesized from programmable logic gates.
High (e.g., 667 MHz).	Moderate/Low (e.g., 100 MHz).
Fixed architecture.	Highly configurable (add/remove FPU, caches etc).
Hard-wired memory controller built-in.	Requires instantiation of a complex MIG IP block for DDR memory communication.
Zero (Sits outside the FPGA fabric).	High (Consumes LUTs, BRAM, and DSPs).
Running operating systems (Linux) and high-speed primary control.	Bare-metal control tasks on pure FPGA boards (Kintex-7).

Hardcore Processor (Zynq ARM)	Softcore Processor (Microblaze)
Complex boot sequence, ARM processor boots first. It implements FSBL(First Stage Boot Loader).	FPGA wakes up and load its bitstream.